Fiscal Year:	FY 2010	**Task Last Updated:**	FY 12/30/2010
PI Name:	Globus, Ruth Ph.D.		
Project Title:	Simulated Microgravity and Radiation-Induced Bone Degeneration: Oxidative Stress- and p53-Dependent Mechanisms		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation Biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Bone Fracture**:Risk of Bone Fracture due to Spaceflight-induced Changes to Bone (2) **Cardiovascular**:Risk of Cardiovascular Adaptations Contributing to Adverse Mission Performance and Health Outcomes (3) **Osteo**:Risk Of Early Onset Osteoporosis Due To Spaceflight		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	650-604-5247
Organization Name:	NASA Ames Research Center		
PI Address 1:	Bone and Signaling Laboratory		
PI Address 2:	Space Biosciences Research Branch		
PI Web Page:			
City:	Moffett Field	**State:**	CA
Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 Radiation Biology NNH04ZUU005N
Start Date:	10/01/2005	**End Date:**	09/30/2010
No. of Post Docs:	4	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:	NOTE: Received NCE to 9/30/2010 per A. Chu/ARC (Ed., 5/2010) NOTE: Received NCE to 3/31/2010 per A. Chu/ARC (Ed., 7/2009) NOTE: Changed Division and Discipline/Program to HRP as of FY2006, per program changes at that time, per JSC/A. Chu-ARC (Ed., 4/2009)		
Key Personnel Changes/Previous PI:	No PI changes. Dr. Joshua Alwood, Dr. Akhilesh Kumar started as full time post-doctoral fellows within this period. An additional postdoctoral fellow, Anuran Chatterjee contributes part time effort.		
COI Name (Institution):	Almeida, Eduardo A.C. (NASA Ames Research Center) Limoli, Charles (University of California Irvine) Searby, Nancy (NASA Ames Research Center)		
Grant/Contract No.:	INTRAMURAL		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	To determine the risks and mechanisms of tissue degeneration in the spaceflight environment, we will examine the combined effects of radiation and simulated weightlessness on bone. We will focus our studies on two important pathways, p53 and oxidative and nitrative stress-related species, in bone marrow progenitor populations because there is abundant evidence that the biological effects of weightlessness and those of radiation share these mechanism. Experiments will be conducted using the rodent hindlimb unloading model, the best validated animal model for simulating weightlessness, followed by irradiation. In Aim 1, we will test the hypothesis that simulated weightlessness of mice modulates the response of bone and bone marrow to radiation, both acutely and during tissue regeneration. In Aim 2 we will identify the target marrow-derived stem, progenitor and differentiated cell populations sensitive to the combined stimuli of simulated weightlessness and radiation. In Aim 3, we will explore the relevance of specific ROS/RNS and p53-dependent mechanisms for mediating cellular responses to combined simulated weightlessness and radiation. The results of these studies are expected to provide important information for the development of countermeasures both to help prevent degeneration of connective tissues such as bone during spaceflight and to improve subsequent tissue regeneration.		
Rationale for HRP Directed Research:			
Research Impact/Earth Benefits:	This project addresses several of the risks identified in NASA's Bioastronautics Critical Path Roadmap (BCPR) caused by long duration spaceflight. We will obtain basic new knowledge about bone tissue degeneration caused by musculoskeletal disuse and radiation at the molecular, cell, and tissue level, relevant to Earth, as well as in space. Results will also lead to the formulation of new countermeasures concepts and risk assessment strategies for bone and connective tissue degenerative diseases.		

Task Progress:	We were the first to report in the scientific literature on the influence of simulated weightlessness and radiation on mineralized tissue. Results obtained in the course of this project are described in five peer-reviewed, scientific papers and two additional manuscripts are in preparation at this time. The outcome is summarized below. Key questions are whether or not low doses and unique types of radiation that astronauts are exposed to in space are detrimental to mineralized tissue, and if space radiation worsens the bone loss that occurs due to unloading of muscles and bones in the reduced gravity environment of space. First, we conducted relatively extensive dose-response and time course experiments using radiation such as high energy Fe-56 (iron), which is encountered in space. Our published results show that total body irradiation of adult male mice with doses of iron as low as 0.10 Gy and doses of photons (Cesium-137) as low as 1.0 Gy can cause significant loss of mineralized tissue within the spongy areas of bone. This bone loss is very rapid (within 3 to 7 days), and appears to be caused by an increase in the activity of bone resorbing osteoclasts. The importance of the osteoclasts for radiation-induced bone loss is further supported by our findings that radiation can increase the number of osteoclasts in cultured cells. When unloading to simulate weightlessness and radiation are combined, changes emerge over time, which are greater than each treatment alone can exert. Mechanical tests are arguably the gold standard for assessing the main function of bone, which is to support the body at rest and in motion. Unloading and radiation each alone have modest effects to lower the yield force in the lumbar spine, defined as the force applied that causes failure, here in compression (Alwood et al. 2010). However, when unloading and radiation are combined, there is a significant reduction in yield force. This combined effect was accompanied by compromised structural integrity of cancellous tissue in the long bones (tibiae) as well as in the spine. One of our surprising findings is that radiation-induced bone loss is restricted to high turnover rate cancellous tissue, in contrast to unloading-induced bone loss, which manifests in both cortical bone tissue (outer shell) as well as cancellous tissue. As a consequence, the mechanical properties of the spine are largely preserved after low dose, total body irradiation alone and are likely to be preserved for most bones, at least in the short term. We also have obtained several lines of evidence in support of the hypothesis that cells responsible for bone formation, osteoblasts and their stem cells and progenitors that reside in bone marrow, may contribute to bone loss caused by radiation. Irradiation can reduce the activity of osteoblasts in cancellous bone (Kondo 2010). In the short term, the combination of unloading and iron has a greater acute effect (-60%) to reduce the number of stem/progenitors in the marrow for osteoblasts than either treatment alone (-20%, -25% respectively) (Yumoto 2010). These results suggest bone marrow stem and progenitor cells for the bone-forming osteoblasts may be more sensitive to adverse effects of radiation under conditions of disuse in space, than occurs under normal conditions on Earth. These findings raise the possibility that radiation may damage stem cells or progenitors for osteoblasts so that recovery of skeletal tissue from the space environment also is impaired. In the short term (1 mo), we found that recovery of the spine from disuse in irradiated mice (0.5Gy iron) was similar to controls (Alwood et al. 2010). Reasoning that if low dose radiation damages stem and osteoprogenitor cells, adverse effects may manifest much later in life as the mice age. To test this possibility, mice were unloaded then exposed to iron, then released to ambulate normally for six months. We found that bone structural changes in mice that had been unloaded (but not irradiated) were consistent with the adaptive changes that occur in astronauts one year after prolonged spaceflight. In contrast, irradiation of unloaded mice with a low dose of iron (0.5Gy) impaired adaptive recovery. These results support these hypothesis that low dose, iron radiation impairs adaptive remodeling of bone due to the damage inflicted upon stem and osteoprogenitor cells that reside within the marrow. We hypothesized that increased oxidative stress caused by the release of unstable reactive molecules, Reactive Oxygen Species (ROS) within the tissue mediates radiation-induced bone loss while disuse activates different mechanisms. Irradiation, but not hindlimb unloading, reduces viability and transiently increases bone marrow cell death (Kondo et al. 2009). Oxidative damage to lipids within mineralized tissue also increases at this time, presumably due to excess ROS (Kondo et al. 2010). Furthermore, a potent anti-oxidant, alpha-lipoic acid, given at the time and following radiation exposure minimizes the rapid bone loss caused by radiation (Kondo et al. 2010). Together, these results show that photon radiation, but not disuse, may increase osteoclasts and acute bone loss via increased oxidative stress. The ability of an anti-oxidant to protect bones from exposure to high energy radiation, such as iron, is still not known. The finding that alpha-lipoic acid protects bone from the detrimental effects of radiation has potential relevance in the future as a dietary countermeasure for astronauts. The question of whether or not weightlessness and/or radiation exert chronic, system-wide effects is an important one. Our results show exposure to space radiation causes persistent oxidative damage to bone marrow proteins in unloaded mice at lower doses of radiation than in normally loaded controls. We propose weightlessness interferes with the ability of cells in bone and marrow to protect from oxidative damage, which leads to persistent oxidative stress, osteoclast-mediated resorption, damage to stem and osteoprogenitors and impaired adaptive remodeling. Although other explanations are possible for our findings, given the well-established effect of radiation to cause oxidative and DNA damage to stem cells and inflammation, we will test this hypothesis in future efforts to improve understanding of spaceflight. The importance of these findings is that simulated weightlessness at the time of radiation exposure appears to impair skeletal remodeling even during a prolonged period of recovery. Furthermore, relatively low doses of space radiation can exert system-wide and persistent change. All of our experiments have been conducted using adult, male mice of a single strain (C57Bl/6J). Differences in biological responses to treatments can occur simply due to age, gender, strain, as well as species, and therefore further studies are needed to help assess human risk. Nonetheless, our findings suggest space radiation, alone and combined with musculoskeletal disuse, may pose a long-term risk to the skeletal health of astronauts. Education and Outreach: In addition to these scientific accomplishments, two Education and Outreach efforts were implemented, consistent with NASA's mission. The first element consisted of mentoring undergraduate and graduate students in our radiation research project. Students from various colleges and graduate training programs have conducted research activities to accomplish the Specific Aims of this grant. The second element engaged a Grade 8-12 web-based project, the Space Settlement Design Contest. This contest has a 16-year track record of success at NASA ARC, inviting students to incorporate radiation protection and radiobiology in their space settlement designs.

Bibliography Type:	Description: (Last Updated: 06/04/2025)
Abstracts for Journals and Proceedings	Globus RK. "Bone progenitor cell responses to irradiation and the regulation of skeletal remodeling." Stem Cell Symposium. 55th Annual Meeting of the Radiation Research Society, Savannah, Georgia, Oct 3-7, 2009. 55th Annual Meeting of the Radiation Research Society, Savannah, Georgia, Oct 3-7, 2009. p. 30. , Oct-2009
Abstracts for Journals and Proceedings	Lee C, Meirer F, Almeida EA, Andrews JC, van der Meulen MCH, Chen J, Globus RK, Mojarrab R, Pianetta P, Alwood JS. "Hard X-Ray nanotomography and quantitative analysis of mineralization in single trabeculae from mice subjected to musculoskeletal disuse and heavy-ion irradiation." SLAC National Accelerator Laboratory. LCLS/SSRL Annual Users' Meeting, Menlo Park,CA. October 18-21, 2009. LCLS/SSRL Annual Users' Meeting, Menlo Park,CA. October 18-21, 2009. , Oct-2009
Abstracts for Journals and Proceedings	Globus RK, Alwood JA, Limoli CL. "Oxidative stress and skeletal health with low dose, ionizing radiation." Low Dose Radiation Research Program Investigators' Workshop, Washington, D.C., April 12-14, 2010. Low Dose Radiation Research Program Investigators' Workshop, Washington, D.C., April 12-14, 2010. , Apr-2010
Abstracts for Journals and Proceedings	Globus RK,Alwood J,Yumoto K, Almeida E, Searby ND, Limoli CL. "HZE and skeletal recovery from simulated weightlessness in mice." 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, Long Island, NY, May 17-19, 2010. 21st Annual NASA Space Radiation Investigators' Workshop, Port Jefferson, Long Island, NY, May 17-19, 2010. , May-2010
Abstracts for Journals and Proceedings	Globus RK. "Vertebrates in Space." Symposium (chair and speaker). NASA ISS Research Academy and Pre-application Meeting in League City, Texas, August 2-4, 2010. NASA ISS Research Academy and Pre-application Meeting in League City, Texas, August 2-4, 2010. , Aug-2010
Abstracts for Journals and Proceedings	Globus RK, Alwood J Yumoto K, Almeida E, Limoli C. "Adverse late HZE effects on marrow progenitors and recovery of bone from disuse." 56th Annual Meeting of the Radiation Research Society, Maui, Hawaii, Sept 25-29, 2010. 56th Annual Meeting of the Radiation Research Society, Maui, Hawaii, Sept 25-29, 2010. http://www.abstractsonline.com/Plan/ViewAbstract.aspx?mID=2569&sKey=359561f3-493c-4d5c-b014-47e677719c70&cKey=440a6c24-b47d-4139-97bc-b37481cd9d0d&mKey=5fb93787-2553-4b3b-a915-8c1ac71f127a , Sep-2010
Articles in Peer-reviewed Journals	Kondo H, Yumoto K, Alwood JS, Mojarrab R, Wang A, Almeida EA, Searby ND, Limoli CL, Globus RK. "Oxidative stress and gamma radiation-induced cancellous bone loss with musculoskeletal disuse." J Appl Physiol. 2010 Jan;108(1):152-61. Epub 2009 Oct 29. PMID: 19875718 , Jan-2010
Articles in Peer-reviewed Journals	Yumoto K, Globus RK, Mojarrab R, Arakaki J, Wang A, Searby ND, Almeida EA, Limoli CL. "Short-term effects of whole-body exposure to (56)fe ions in combination with musculoskeletal disuse on bone cells." Radiation Research, 2010 Apr;173(4):494-504. PMID: 20334522 , Apr-2010
Articles in Peer-reviewed Journals	Alwood JS, Yumoto K, Mojarrab R, Limoli CL, Almeida EA, Searby ND, Globus RK "Heavy ion irradiation and unloading effects on mouse lumbar vertebral microarchitecture, mechanical properties and tissue stresses." Bone. 2010 Aug;47(2):248-55. Epub 2010 May 11. PMID: 20466089 , Aug-2010
Articles in Peer-reviewed Journals	Globus RK, Morey-Holton ER. "Advances in understanding the skeletal biology of spaceflight." Gravitational and Space Biology 2009 Sep;22(2):3-12. , Sep-2009
Articles in Peer-reviewed Journals	Kumar A, Tahimic CGT, Almeida EAC, Globus RK. "Spaceflight modulates the expression of key oxidative stress and cell cycle related genes in heart." Int J Mol Sci. 2021 Aug 23;22(16):9088. https://doi.org/10.3390/ijms22169088 ; PMID: 34445793; PMCID: PMC8396460 , Aug-2021
Books/Book Chapters	Globus RK, Caiozzo VJ, Acharya M, Fike JR, Limoli CL. "Redox regulation of stem cell compartments: The convergence of radiation-induced normal tissue damage and oxidative stress." in "Oxidative Stress in Cancer Biology and Therapy." Ed. D.R. Spitz, K.J. Dornfeld, K. Krishnan, D. Gius. New York : Springer, 2012. p.169-192., Jan-2012